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Modeling and Analysis of HV Cable Ampacity for Power Flow Optimization

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Abstract—This paper presents a new approach for the determination of underground cable ampacity that considers surrounding medium of the cable, and compares the results from the FEM model with standard approach. It shows how ampacity of a buried cable is affected by the extent of heat transfer from the cable to the surrounding soil and also by the heterogeneity of the soil and its thermal characteristics. Numerical and finite element model of steady-state thermal analysis and ampacity evaluation are presented in this paper. COMSOL software is used for the three dimensional simulation of a 44kV armored HVAC XLPE cable buried directly in native soil. The methodology includes mathematical solutions for heat transfer equations to calculate and obtain the temperature at the cable surface and results show the optimal acceptable ampacity. The results from the proposed method provide solutions for ampacity problems that require flexibility and dynamic approach for real life scenarios that are not present in the literature and previous works.

Index Terms—Thermal field, Finite element method, cable ampacity, underground cable system, power flow

I. INTRODUCTION

Ampacity calculation of a cable is complicated particularly in cases where the routing goes through various environments with dynamic thermal conditions. The heterogeneous nature of the soil and existence of heat transfer by convection in water and air requires a two dimensional heat transfer approach to the problem. Temperature distribution in the cable and surrounding is a function of variable soil thermal resistance, location from the ground level along the axis and depth of burial, radial distance between nearby cables etc., [1]. An example cable installation is simulated to compare our results with the standard tables [2]–[7]. The results are comparable only with homogeneous soil; as standards do not consider heterogeneous nature, varying media and natural convection, surface-to-surface radiation in the participating media.

Though ampacity calculation follows traditional numerical methods presented in the literatures and standards, very few works have been presented for the non-uniform nature of the medium surrounding the cables. This includes models presented by the pioneers Neher- McGrath [8], however these are directly applicable to underground cables that are installed in homogeneous environment. The thermal model developed by them is based on many assumptions to make the numerical formula simpler and in that approach one of the assumptions

is the uniform surrounding medium. Hence, no geometrical and thermal property changes along the cable was considered that makes the approach not anywhere close to reality. Hence the problem is reduced to two-dimensional problem from a three-dimensional one. In addition, the problem was further simplified by considering a virtual heat sink above the ground surface where the cable is installed. Hence, the heat transfer problem becomes a one-dimensional numerical formulation and the ampacity formulation was solved directly as an algebraic equation [8].

Most of the accepted standards [2]–[7] follow the approximate model provided in the Neher-McGrath method. These standard table hence follow the same homogeneity principle and does not account for any changes in the characteristics of the surrounding medium along the cable length that directly affects the cable ampacity [9], these non-uniform conditions are more prevalent in urban areas and in cases where the cable passes through paved parking lots with concrete bases, the heat dissipation will be poor thus increasing the cable internal temperatures [10]. In the work presented in [11], assessment methods to calculate ampacity for underground power cables are shown. It is understood that all the above approaches consider homogeneous environment for the surrounding medium thus limiting the validity of the results. In more conservative approaches and in situations where complex heterogeneous conditions prevail for longer sectors along the cable length, the worst-case approach is followed. Cables are considered to be installed in the worst thermal environment and cable ampacity derating is done keeping this a base [12].

No clear direct method or guide is standardized for cable ampacity calculation for small heterogeneous portions having poor thermal properties prevail around the cable [9]–[13]. This indicates that the cable ampacity values are not constant as defined in the standard tables and in some cases deviations are present. The cables subjected to continuous operations will lead to changes in the ambient environment which will also affect the heat transfer ability of the surrounding and in turn leads to cable damage. Many new methods for estimating power cable thermal performances in underground installation conditions are presented in [14]–[16]. They follow finite element analysis for power cables in homogeneous soil installed in conduits [9]–[15].

This paper discusses in details the conventional ampacity estimation and using FEM ampacity corrections with temperature distribution for single core armored power cables buried directly in the soil in flat formation and trefoil fashion. The soil homogeneous and heterogeneous conditions are also incorporated in the results. The temperature of a cable conductor will reach its maximum allowable value when it is loaded with the rated load [17]. The heat generated by the cable conductor core is dissipated to the surrounding environment via the conductor screen, insulation layer and armor and/or the cable sheath. The relation between the temperature at the conductor core and the sheath will be used to calculate the true ampacity of the cable.

The heat generated in the cable is from the losses. In addition to the losses resulting from current flow, other losses in the cable that also result in heat generation are the dielectric losses, sheath, screen and armor loss factors. The IEC 60287 standard tables [3], [5] are used as a baseline to determine the cable component thermal resistances and numerical ampacity calculation. The FEM model is used to correct the ampacity derived from standard calculations.

The paper is structured in the following ways. Section II gives the basis of the finite element model equations and cable conductor temperature equations that are used in background of FEM. The model simulation and results for varying the cable surroundings are elaborated in section III. The conclusion and discussion are provided in section IV.

II. FINITE ELEMENT MODEL

The FEM model aims to illustrate the heat dissipation in the cable and surrounding soil. The cable under discussion is considered to be buried at least 1 meter deep in a homogeneous soil with density 1500 kg/m³. The cable length is far larger than its diameter. The copper core conductor is shielded by extruded thermos-set semi-conducting compound. The insulation is cross-linked polyethylene (XLPE) and steel armoring is used, compounded jute and fibrous material is chosen for armor serving [2]. The cable cross section view is shown in Fig.1. The design parameters for the cable and surrounding is given in Table I. Two types of cable laying techniques are investigated for the 3-phase single core cable in the paper - flat formation and trefoil arrangement. The surrounding soil medium is modeled as a huge cylindrical structure of uniform density around the length of the cable.

A. Heat Transfer Basics

To determine cable ampacity and heat distribution study, finite element modeling software COMSOL with finer mesh structure is used. The heat transfer problem is solved by calculating field temperature with given heat source and rate of heat transfer [18]. Here we solve for the temperature of cable conductor given the heat source. The heat resulting from the cable losses and the heat input rate for the conductor are used to determine maximum allowable temperature for the cable core. Approximate current values are used from standard tables to start with, then temperature distribution is calculated iteratively until the cable temperature reaches the desired value

i.e. target temperature for copper conductor is 90°C., using the cable ampacity equation.

TABLE I
CABLE AND INSTALLATION DATA

Parameter	Value
Voltage (kV)	44
Conductor Area (sq.mm)	107
Conductor diameter (mm)	61
Conductor	Copper
Conductor Temperature (°C)	90
Loading Factor	1
Insulation	XLPE
Insulation Thickness (mm)	11.6
Aarmor	Steel Wires
Aarmor Thickness(mm)	5.16
Aarmor Serving	Compounded Jute Fibre
Duct Type	Direct Buried
Aambient temperature (°C)	20
Soil Resistance (°C.m/W)	0.9
Frequency (Hz)	50

The proposed approach follows a finite element model with finer mesh generated and the heat transfer equations are solved for the temperature at the cable surrounding [18].

The soil surface is considered isothermal, soil resistivity is an important factor because it determines the heat dissipation capability of the medium touching the cable surface while the resistance of air on soil surface is neglected as it has no effect on temperature distribution [1]. Infinite boundary conditions are used and the heat source is the energy produced from the internal heating of the cable conductor due to the current flow [1].

44 kV XLPE insulated single core (copper) cable is modeled using the FEM. The conductor core temperature is determined from the cable losses as a result of rated ampacity.

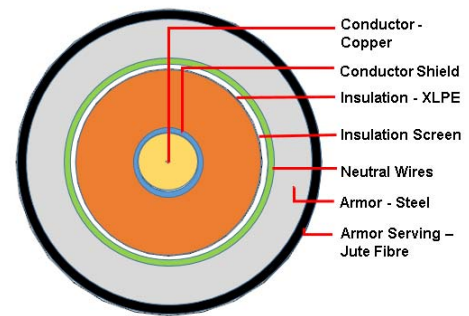


Fig. 1. Cable Cross Section - showing Cable Structure

The thermal flow of the cable with single core is given by

$$\frac{\lambda_t}{k_i} \left(\frac{\partial^2 T}{\partial x^2} + \frac{1}{x} \frac{\partial T}{\partial x} \right) + \frac{W_i}{k_i} = \frac{\partial T}{\partial t}, \quad (1)$$

where, λ_t - thermal conduction coefficient; k_i thermal capacity of insulation; W_i thermal flow; T temperature; t - time.

B. Ampacity Calculation

By resolving the differential equation, the relation between the core (T_c) and sheath temperature (T_s) in a single core buried cable can be given as,

$$T_c - T_s = W_c\{(T_1 + (1 + \lambda_1)T_1) + (1 + \lambda_1 + \lambda_2)(T_3 + T_4)\} + W_d\{0.5T_1 + T_2 + T_3 + T_4\}, \quad (2)$$

W_c and W_d are conductor loss and dielectric loss per unit length of the cable, T_1 is the thermal resistance between conductor and sheath, T_2 is the thermal resistance between sheath and armor, T_3 is the thermal resistance of the outer covering. T_4 is the thermal resistance of surrounding medium (ratio of cable surface temperature rise above ambient to the losses per unit length). λ_1 and λ_2 are the sheath and armor ratio compared to the conductor losses. The ambient temperature for the FEM is considered to be 20°C and the conductor size is 107 mm².

Ampacity for AC cables is

$$I = \sqrt{\frac{\Delta\theta - W_d[0.5T_1 + n(T_2 + T_3 + T_4)] + (v - 1)\Delta\theta_x}{R[T_1 + n(1 + \lambda_1)T_2 + n(1 + \lambda_1 + \lambda_2)(T_3 + T_4)]}}, \quad (3)$$

Where

$\Delta\theta$ is the conductor temperature rise above the ambient temperature (°C);

R is the a.c. resistance of the conductor at its maximum operating temperature (ohms/m);

v is the ratio of the thermal resistivities of the dry and moist soil zones ($v = \rho_d/\rho_w$);

ρ_d is the thermal resistivity of the dry soil (°Cm/W);

ρ_w is the thermal resistivity of the moist soil (°Cm/W);

θ_x is the critical temperature of the soil and temperature of the boundary between dry and moist zones (°C);

θ_a is the ambient temperature (°C);

$\Delta\theta_x$ is the critical temperature rise of the soil. This is the temperature rise of the boundary between the dry and moist zones above the ambient temperature of the soil ($\theta_x - \theta_a$) (°C);

IEC standard 60287 [5] is used for determining the thermal resistances and losses.

$$T_1 = \frac{\rho_T}{2\pi} \ln\left[1 + \frac{2t_1}{d_c}\right], \quad (4)$$

where ρ_T is the thermal resistivity of insulation (°Cm/W); t -time.

d_c is the diameter of conductor (mm);

t_1 is the thickness of insulation between conductor and sheath (mm).

Similarly T_2 and T_3 are calculated for the cable. The cable core temperature calculated for an ampacity (from standard

procedures) of 344 A is 90°C.

The cable conductor is the heat source. The heat transfer in solids occurs in three mechanisms- conduction, convection and radiation. Since the cables are directly buried and no convection heat transfer is expected. The heat transfer by conduction equation of cables directly buried in the soil is given by

$$-\nabla (K\nabla T) = Q + h(T_a - T_c), \quad (5)$$

k is thermal conductivity, h is the heat transfer coefficient and q is the heat source. The core conductor serves as the heat source and the generated heat is the resultant of the joule heating obtained from equation 5. The heat from convection and radiation is more relevant for cables in air medium.

III. ANALYSIS FROM SIMULATION

The simulation is carried out in steady-state conditions in homogeneous and heterogeneous native soil surroundings. The cables laying style is shown in Fig.2 in flat formation are separated with a distance of 150 mm and cables in trefoil arrangement are touching each other. The COMSOL Multi physics software was used for generating finer free tetrahedral mesh with 1,263,641 elements as shown in Fig.2

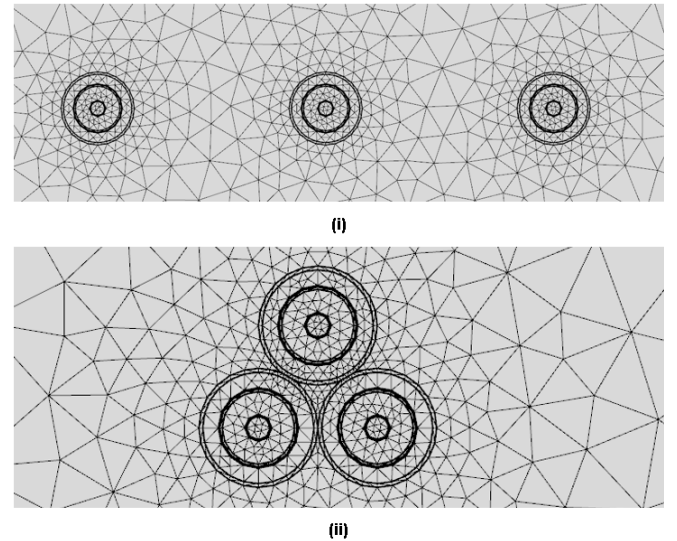


Fig. 2. Cable Arrangement for Study with Fine Mesh (i) Flat-formation (ii) Trefoil

The cable total losses (W) owing to the cable conductor losses and dielectric losses are given by,

$$W = W_{c'} + W_d, \quad (6)$$

$W_{c'}$ is the cable internal losses due to conductor loss W_c from the current flowing through the conductor (I) and conductor AC resistance (R), sheath and armor losses (λ_1, λ_2) and W_d is the dielectric loss of the insulation.

$$W_{c'} = I^2 R(1 + \lambda_1 + \lambda_2), \quad (7)$$

$$W_d = \omega C U_0^2 \tan \delta, \quad (8)$$

where, $\omega = 2\pi f$, f is the AC current frequency, C is the electrical capacitance, U_0 phase-ground voltage. The capacitance is obtained from the standard calculation used in IEC standard found in [3]. $\tan \delta$ is the dielectric loss factor of the insulation and it is 0.004 for the XLPE material.

A. Homogeneous Soil Results

A three phase HVAC cable was constructed in the finite element model using COMSOL multiphysics software. The initial assumption is homogeneous soil environment is surrounding the cable with thermal resistivity of 0.9°C.m/w and the cable is considered to be buried directly in the soil at a depth of 1 meter from the ground level. The ampacity calculated using standard procedure [3] and results from FEM simulation are discussed below. The results show the validity of the FEM simulation as the ampacity values for ideal conditions are close to the standard procedure. When the FEM model is used to obtain the steady state condition, the ampacity decreases to 315 A due to the change in the local ambient temperature change arising from the change in soil thermal resistivity. The FEM model is instrumental to find the critical temperature rise limit in the local ambient from the continuous operation of the cable resulting in the change in thermal resistance of the surrounding medium. Fig.3 shows the temperature distribution of the cable and its surrounding. The soil critical temperature rise is 62°C . To prevent further changes in the soil thermal resistivity, the new ampacity is calculated. It is clearly seen that the temperature rise for the flat formation is less compared to the trefoil formation, because in trefoil arrangement the adjacent cables touching each other raises the sheath temperature due to internal conduction.

TABLE II
AMPACITY COMPARISON FOR HOMOGENEOUS SOIL

Type of Installation	Standard Calculation / Dry-out	Simulation / Dry-out
Cables in Flat Formation	344A/338A	347.24A/340A
Cables in Trefoil Formation	295A/288A	299A/290.2A

B. Heterogeneous Soil Results

Cable ampacity calculation for heterogeneous soil condition is not included in standard procedures. Generally, the cable base ampacity for the homogeneous soil condition is de-rated for the worst case scenario and according to [16] arbitrarily average of the soil thermal resistivities is considered. But in this approach for the standard calculation we consider the cable ampacity for the worst soil thermal resistivity i.e 1.6°C.m/W .

To investigate the effect of heterogeneous soil, the soil is divided into 2 sections with different thermal resistivity (0.9 and 1.6°C.m/W). The soil segments are divided along the length of the cable. These scenarios occur in cable installations that pass through different soil regions. Each segment is

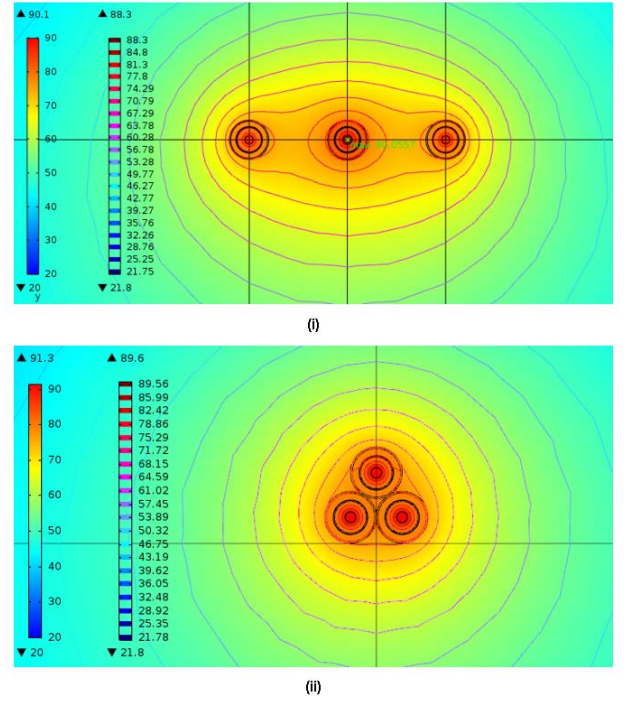


Fig. 3. Temperature Distribution for Cables in Homogeneous Soil (i) Flat-formation (ii) Trefoil

considered to have uniform density. In this simulation, cables installation position is same as the homogeneous section with same boundary conditions and loading factor. Fig.4 shows the view of the temperature contours along the length for 3-phase cables. Temperature distribution is symmetrical and the heat transfer is higher in higher conductive region. The contours show that the surface temperature for the higher resistive soil is slightly higher than the other section which is observable at the cable surface near the section division in Fig.4. Table III compares values achieved for the heterogeneous soil with standard derated ampacity for the worst-case thermal resistivity.

TABLE III
AMPACITY COMPARISON FOR HETEROGENEOUS SOIL

Type of Installation	Standard Calculation	Simulation
Cables in Flat Formation	279A	308A
Cables in Trefoil Formation	239A	263A

As conservative approach was followed in the standards for cable derating and no standard approach for heterogeneous soil condition, according to [16], T_4 can be derived from the FEM but in the paper it is considered that θ_a as 0 and the cables losses are assumed to unity. But since the losses and ambient temperature are real, the FEM simulation data are used to re-calculate the cable ampacity for heterogeneous soil. The temperature rise at the surface of the cable and its relation with T_4 is given by [16] in the below equation and it compares well with the standards.

$$\theta_s - \theta_a = T_4 W, \quad (9)$$

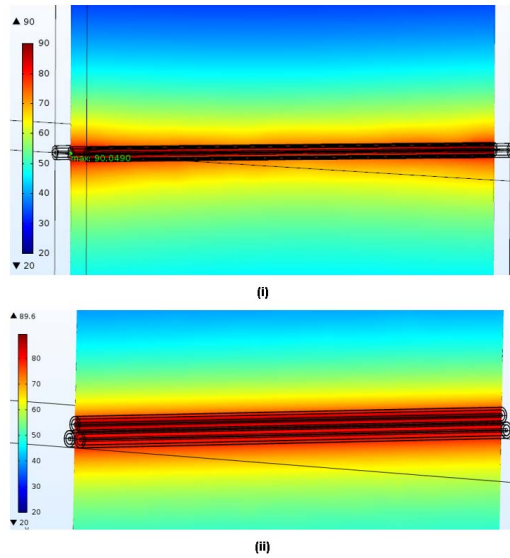


Fig. 4. Temperature Distribution for Cables in Heterogeneous Soil Along the Length of the Cable (i) Flat-formation (ii) Trefoil

Here, θ_a is 20°C and the total cable losses calculated from equation 6. The heat distribution in the heterogeneous condition is non-uniform but the heat conduction is towards the thermally favorable region with lower thermal resistance [18], hence the heat is not trapped in the medium but the heat exchange rate is slower. Hence we can consider the average temperature instead of the maximum temperature for the surface. The same approach is followed for de-rating for the soil thermal resistance changes from continuous operation. We can see an improvement in the cable ampacity in Table IV with the above approach.

TABLE IV
AMPACITY COMPARISON FOR HETEROGENEOUS SOIL WITH AVERAGE TEMPERATURE APPROACH

Type of Installation	Standard Calculation	Average Temp Approach	Simulation (Dry-out)
Cables in Flat Formation	279A	3084A	301.2A
Cables in Trefoil Formation	239A	263A	256.6A

IV. CONCLUSION

The ampacity of the power cables depends on the thermal properties of the cable surrounding environment. Due to cable continuous operation a change in thermal characteristics of the medium is observed. By heat transfer model using FEM, the results show that limiting the soil dry-out region to a set value is possible and it is observable from the temperature distribution contours. For homogeneous and heterogeneous soil, by using average temperature approach and temperature contours for dry soil band an improved ampacity from standard calculations are observed. Any change in the surrounding medium changes the cable internal heating and in turn results in ampacity correction.

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REFERENCES

- [1] A. Mahmoudi, S. Kahourzade, and D. S. S. Lee, "Cable ampacity calculation in heterogeneous soil using finite element analysis," in *2011 5th International Power Engineering and Optimization Conference*, Jun. 2011, pp. 416–421.
- [2] "IEEE Standard Power Cable Ampacity Tables." IEEE N.Y. Std., 835, 1994.
- [3] "IEC Standard Electric Cables-Calculation of Current Rating-Part 1: Current Rating equations (100% load Factor) and calculation of Losses-Section 1: General." IEC Std., 287, 2015.
- [4] "IEC Standard Electric Cables-Calculation of Current Rating-Part 1: Current Rating equations (100% load Factor) and calculation of Losses-Section 2 Sheath Eddy Current Loss Factors for Two Circuits in Flat Formation." IEC Std., 287, 1994.
- [5] "IEC Standard Electric Cables-Calculation of Current Rating-Part 2: Thermal resistance-Section 1: Calculation of thermal resistance." IEC Std. 287, 1994.
- [6] "IEC Standard - A Method for Calculating Reduction Factors for Groups of Cables in Free Air, Electric Cables- Calculation of the Current Rating-Part 2: Thermal Resistance Section 2A: Protected from Solar Radiation." IEC Std., 287, 1995.
- [7] "IEC Standard. Electric Cables- Calculation of the Current Rating-Part 3: Sections on Operating Conditions Section 1: Reference Operating Conditions and Selection of Cable Type." IEC Std. 287-1995., 1995.
- [8] J. H. Neher and M. H. McGrath, "The calculation of the temperature rise and load capability of cable systems," *Power Apparatus and Systems, Part III. AIEE Transactions*, vol. 76, pp. 752–764, 1957.
- [9] C. Hwang, J. Chang, and H. Chen, "Calculation of ampacities for cables in trays using finite elements," *Electric Power Systems Research*, vol. 54, no. 2, pp. 75 – 81, 2000.
- [10] H. Brakelmann and G. Anders, "Ampacity reduction factors for cables crossing thermally unfavorable regions," *IEEE Transactions on Power Delivery*, vol. 16, no. 4, pp. 444–448, Oct 2001.
- [11] F. Aras, C. Oysu, and G. Yilmaz, "An assessment of the methods for calculating ampacity of underground power cables," *Electric Power Components and Systems*, vol. 33, no. 12, pp. 1385–1402, 2005.
- [12] J. Nahman and M. Tanaskovic, "Determination of the current carrying capacity of cables using the finite element method," *Electric Power Systems Research*, vol. 61, no. 2, pp. 109 – 117, 2002.
- [13] C.-C. Hwang and Y.-H. Jiang, "Extensions to the finite element method for thermal analysis of underground cable systems," *Electric Power Systems Research*, vol. 64, no. 2, pp. 159 – 164, 2003.
- [14] P. Vaucheret, R. A. Hartlein, and W. Z. Black, "Ampacity derating factors for cables buried in short segments of conduit," *IEEE Transactions on Power Delivery*, vol. 20, no. 2, pp. 560–565, April 2005.
- [15] M. Al-Saud, M. El-Kady, and R. Findlay, "A new approach to underground cable performance assessment," *Electric Power Systems Research*, vol. 78, no. 5, pp. 907 – 918, 2008.
- [16] F. de Leon and G. J. Anders, "Effects of backfilling on cable ampacity analyzed with the finite element method," *IEEE Transactions on Power Delivery*, vol. 23, no. 2, pp. 537–543, April 2008.
- [17] W. Yang, Y. Liu, Y. Xu, X. Cao, J. Li, and M. Xu, "A method of temperature calculation of power cable conductor," in *2008 International Symposium on Electrical Insulating Materials (ISEIM 2008)*, Sept 2008, pp. 460–462.
- [18] S. Kahourzade, A. Mahmoudi, B. N. Taj, and O. Palizban, "Ampacity calculation of the underground power cables in voluntary conditions by finite element method," in *The 8th Electrical Engineering/ Electronics, Computer, Telecommunications and Information Technology (ECTI) Association of Thailand - Conference 2011*, May 2011, pp. 657–660.